

Diabetic Retinopathy

Introduction

Diabetic retinopathy is a serious eye complication of diabetes that damages the small blood vessels of the retina, the light-sensitive tissue at the back of the eye that is essential for vision.[cite:1] [cite:2] It is one of the leading causes of preventable vision loss and blindness among working-age adults, and its importance has increased as diabetes has become more common worldwide.[cite:1] [cite:4]

The disease develops gradually when persistently high blood glucose levels injure retinal blood vessels, causing them to leak fluid or blood, become blocked, or stimulate the growth of abnormal new vessels.[cite:2] [cite:3] In the early phase, many people do not notice any symptoms, which makes regular eye examinations especially important for timely diagnosis and treatment.[cite:2] [cite:4]

Diabetic retinopathy is broadly divided into non-proliferative diabetic retinopathy and proliferative diabetic retinopathy.[cite:1] [cite:3] In non-proliferative disease, retinal blood vessels weaken and leak, while in proliferative disease, abnormal new blood vessels form in response to retinal ischemia, or inadequate blood supply.[cite:1] [cite:3]

The risk of developing diabetic retinopathy increases with longer duration of diabetes, poor glycemic control, high blood pressure, kidney disease, and other metabolic risk factors.[cite:2] [cite:4] Because early damage may not affect vision immediately, diabetic retinopathy is considered a major public health challenge that requires awareness, screening, and long-term diabetes management.[cite:1] [cite:4]

Stages

Diabetic retinopathy usually progresses through four recognized stages, beginning with mild retinal vessel changes and advancing to severe vision-threatening disease if left untreated. [cite:13] [cite:15] The first three stages are grouped under non-proliferative diabetic retinopathy, while the last stage is proliferative diabetic retinopathy.[cite:13] [cite:21]

Mild Non-Proliferative Diabetic Retinopathy

This is the earliest stage of the disease and is characterized mainly by microaneurysms, which are small bulges in the retinal blood vessels.[cite:13] [cite:15] At this stage, vision may still appear normal, and the changes are often detected only during retinal examination.[cite:13] [cite:18]

Moderate Non-Proliferative Diabetic Retinopathy

In the moderate stage, blood vessel damage becomes more pronounced, and some retinal vessels may start to swell, leak, or become blocked.[cite:14][cite:15] Fluid leakage can affect the macula, leading to blurred central vision in some patients.[cite:14][cite:18]

Severe Non-Proliferative Diabetic Retinopathy

This stage involves widespread blockage of retinal blood vessels, reducing oxygen supply to the retina.[cite:15][cite:17] Clinical signs may include multiple retinal hemorrhages, venous abnormalities, and cotton wool spots, indicating significant retinal ischemia.[cite:15][cite:17]

Proliferative Diabetic Retinopathy

This is the most advanced stage, in which the retina responds to poor blood supply by producing abnormal new blood vessels, a process called neovascularization.[cite:15][cite:17] These fragile vessels can bleed into the vitreous, cause scar tissue formation, and lead to tractional retinal detachment or severe vision loss.[cite:14][cite:16]

Precautions

Precautions for diabetic retinopathy focus mainly on preventing onset, delaying progression, and reducing the risk of vision-threatening complications.[cite:24][cite:31] The most important precaution is maintaining good control of diabetes through regular monitoring of blood sugar and adherence to prescribed treatment.[cite:24][cite:31]

Blood pressure and cholesterol should also be carefully controlled because hypertension and dyslipidemia can worsen retinal vessel damage.[cite:25][cite:29] Regular medical follow-up helps ensure that these systemic risk factors remain within safe ranges.[cite:31]

Routine dilated eye examinations are essential, even when vision seems normal, because diabetic retinopathy may remain silent in its early phases.[cite:26][cite:28] Early identification allows treatment before permanent retinal damage develops.[cite:28][cite:31]

Healthy habits also play a protective role, including balanced diet, regular physical activity, avoiding smoking, and limiting alcohol consumption.[cite:27][cite:30] Eye protection from ultraviolet exposure and injury may also support overall ocular health.[cite:26]

Symptoms

Diabetic retinopathy often causes no obvious symptoms at first, especially during the early stages.[cite:34][cite:37] As the disease progresses, patients may begin to notice visual disturbances caused by retinal swelling, bleeding, or reduced retinal function.[cite:35][cite:36]

Common symptoms include blurred vision, fluctuating vision, floaters, difficulty seeing colors clearly, and dark or empty areas in the visual field.[cite:34][cite:37] Some people also report problems with night vision or distortion in central vision when the macula becomes affected.[cite:37][cite:42]

In advanced cases, sudden vision loss may occur due to vitreous hemorrhage or retinal detachment.[cite:34][cite:8] Because symptoms often appear late, screening remains more reliable than waiting for visual complaints alone.[cite:36][cite:40]

Treatment by Stage

There is no single universal cure that completely removes diabetic retinopathy in every patient, but effective treatment can slow progression, preserve vision, and in some cases improve retinal changes.[cite:34][cite:44] Treatment depends on the stage of the disease and whether diabetic macular edema or proliferative changes are present.[cite:37][cite:44]

Mild Non-Proliferative Stage

At the mild stage, treatment usually focuses on observation and strict control of blood sugar, blood pressure, and cholesterol rather than immediate eye procedures.[cite:21][cite:45] Regular follow-up examinations are advised to detect progression at an early point.[cite:21][cite:34]

Moderate Non-Proliferative Stage

At the moderate stage, closer follow-up is required, and treatment may be added if macular edema develops.[cite:21][cite:45] Anti-VEGF injections may be used to reduce retinal swelling and leakage in selected cases.[cite:44][cite:47]

Severe Non-Proliferative Stage

In severe non-proliferative disease, ophthalmologists may consider anti-VEGF therapy or laser treatment to lower the risk of progression to proliferative diabetic retinopathy.[cite:21][cite:45] Intensive metabolic control remains essential at this stage.[cite:34][cite:44]

Proliferative Stage

Proliferative diabetic retinopathy usually requires urgent specialist treatment because of the high risk of bleeding and retinal detachment.[cite:37][cite:43] Common treatments include anti-VEGF injections, panretinal photocoagulation laser therapy, steroid treatment in selected cases, and vitrectomy surgery when there is non-clearing vitreous hemorrhage or tractional retinal detachment.[cite:43][cite:44][cite:47]

What a Patient Should Do

A person suffering from diabetic retinopathy should continue diabetes treatment seriously and monitor blood glucose regularly, because stable sugar control lowers the chance of worsening eye damage.[cite:24][cite:31] Prescribed medicines for diabetes, blood pressure, and cholesterol should be taken consistently.[cite:25][cite:31]

Eye appointments should never be skipped, and any sudden increase in floaters, blurred vision, flashes, or partial vision loss should be reported immediately to an eye specialist.[cite:34][cite:37] Early medical response can prevent severe complications.[cite:37][cite:44]

Daily lifestyle also matters. Patients should eat a balanced diet, exercise safely as advised by their doctor, avoid smoking, limit alcohol, and protect the eyes from injury.[cite:27][cite:30] In advanced proliferative disease, heavy lifting or very strenuous activity may need to be avoided if advised by the treating ophthalmologist because of bleeding risk.[cite:46][cite:47]